
VERITAS

AI Contract Review for German Freelancers

Catch risky clauses before you sign them.

Business Case & Technical Architecture Report

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Abstract

Veritas is a production-grade AI contract review platform that targets a clear and underserved market: the 1.47 million self-employed professionals in Germany who sign legally complex freelance contracts without access to affordable expert review. A specialised *Fachanwalt* charges €300–600 per hour; most freelancers therefore sign without review and absorb the legal and economic risk silently. Veritas reduces this to a fixed €9.99 monthly subscription with a measurable variable cost of approximately €0.012 per contract analysed.

The technical contribution is a retrieval-augmented three-layer architecture in which the large language model is treated as a swappable language engine, not as a source of legal knowledge. A curated 66-entry playbook of risky clause patterns — each tied to a specific German statute and indexed by text-embedding-3-small in Firestore Vector Search — combined with a normalised database of hourly-rate benchmarks and statutory references, provides the factual grounding that the LLM synthesises into clause-level findings. Every finding the system produces is traceable to a specific row in the knowledge base, which makes the product auditable and structurally resistant to hallucination in a way that prompting GPT-4o directly cannot match.

The deployed system has been evaluated end-to-end on a hand-curated gold set of 50 clauses (v1.2) across twenty-five clause-type categories. The current model achieves a composite precision of 80.7%, with risk-classification accuracy at 76% and statute-citation hit rate at 82%. We discuss the four reinforcing moats that protect the business — a proprietary data flywheel, relationship-led distribution through the VGSD and Steuerberater channels, DSGVO-compliant EU-hosted infrastructure, and user-level lock-in through longitudinal contract history — and argue that none of them can be replicated by a foundation-model release. The report concludes with a roadmap to break-even at approximately 500 Pro subscribers and 50 Team firms.

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1 Introduction

1.1 Motivation

The German freelance labour market sits at the intersection of three trends that make it structurally underserved by existing legal services. First, the segment is large and growing: [Statista \(2023\)](#) count 1.47 million self-employed professionals in liberal-profession categories, and [Bitkom e.V. \(2024\)](#) report that German firms spent €51.7 billion on IT services alone in 2024, much of it flowing through individual freelance contracts. Second, the contracts these freelancers sign are legally dense — typically fifteen to twenty pages of German prose spanning clauses on intellectual property, liability, payment terms, non-compete restrictions, data handling, and the disguised-employment criteria of § 7 SGB IV ([Bundesministerium der Justiz, 2024b](#)). Third, the economic asymmetry between freelancer and counterparty is severe: a specialised *Fachanwalt für Arbeitsrecht* charges €300–600 per hour, and a thorough contract review typically takes one to two hours of billable time. For a freelancer whose entire project engagement is worth €5,000–15,000, spending €600 on legal review for each contract is economically irrational. The result is that the vast majority of freelancers sign contracts without any professional review at all, absorbing risks they often do not even recognise.

Generic large language models do not solve this problem. ChatGPT and comparable tools hallucinate German statutory references — the Stanford RegLab study by [Dahl et al. \(2024\)](#) documents hallucination rates of 58–82 % on legal queries across leading LLMs, with follow-up work ([Magesh et al., 2025](#)) showing that even commercial retrieval-augmented legal research tools hallucinate roughly one query in six. A freelancer pasting their contract into ChatGPT also commits a DSGVO-relevant data export to a US-hosted inference endpoint, which is increasingly out of step with post-Schrems-II ([Court of Justice of the European Union, 2020](#)) compliance requirements and frequently in breach of NDA clauses inside the very contract being reviewed.

1.2 Contribution

This report describes Veritas, a deployed end-to-end system that addresses the gap. The principal contributions are:

- A three-layer retrieval-augmented architecture in which the LLM is restricted to synthesis of pre-retrieved facts. Statutes are looked up from a curated reference table; hourly-rate benchmarks come from a normalised database derived from [freelancermap GmbH \(2025\)](#); and risky-clause patterns are matched via vector similarity against a hand-curated 66-entry playbook. The model is permitted to write German legal prose, but never to invent a §-number, a case citation, or a benchmark figure.
- An on-device PII-redaction layer using [Microsoft \(2023\)](#) and [Honnibal et al. \(2020\)](#) which strips names, organisations, locations, and IBANs from extracted contract text before any external API call. Combined with EU-hosted infrastructure (Google Cloud Run, `europa-west1`), this enables a DSGVO-compliant data flow that excludes general-purpose

AI tools from the regulated-vertical segment entirely.

- A formal evaluation harness that scores the analyser on a hand-curated gold set of 50 risky clauses (v1.2) across twenty-five clause-type categories, with three orthogonal metrics combined into a single composite precision score. The current production model attains $P = 0.807$.
- A business model with verified unit economics: variable inference cost is approximately €0.012 per contract, against an effective Pro-tier price of €0.999 per contract; gross margin is approximately 98.8%, and break-even is reached at roughly 500 paying subscribers.

1.3 Structure of the report

Section 2 surveys the relevant literature on retrieval-augmented generation, legal language models, and hallucination mitigation. Section 3 formalises the problem and Section 4 sizes the market. Section 5 describes the user-facing product. Sections 6–8 detail the system architecture, the AI pipeline, and the construction of the knowledge base. Section 9 covers hallucination and privacy safeguards. Sections 10–11 present the evaluation methodology and results. Section 12 returns to the business case — go-to-market, pricing, and corrected unit economics. Section 13 discusses defensibility. Sections 14 and 15 close with limitations and the roadmap to break-even.

2 Related Work

2.1 Retrieval-augmented generation

The architectural decision to ground every factual claim in a retrieved database row rather than in model weights follows directly from the retrieval-augmented generation (RAG) framework introduced by Lewis et al. (2020), which combines a parametric language model with a non-parametric memory of indexed documents. The motivating argument is identical: parametric memory is frozen at training time and cannot be updated without retraining, whereas a vector index can be amended in hours. For legal applications this distinction is decisive. German labour-law jurisprudence is actively evolving — the Bundessozialgericht (2024) ruling in April 2024 (B 12 BA 9/22 R) materially tightened the disguised-employment test — and any system whose legal knowledge is baked into model weights will silently drift out of compliance with current case law.

Subsequent work on grounded generation has consistently confirmed that retrieval reduces hallucination on knowledge-intensive tasks (Ji et al., 2023), though it does not eliminate it: the model can still misattribute, conflate, or selectively cite the retrieved context. This residual risk is what motivates the additional architectural constraints described in Section 9, in particular the verbatim-quoting rule and the statute-lookup-not-generation rule.

2.2 Legal language models and contract review

The legal NLP literature offers two complementary lines of work. *Domain-pretrained encoders* such as Legal-BERT (Chalkidis et al., 2020) produce embeddings that outperform general-

purpose encoders on legal classification benchmarks; however, they are not generative and cannot produce the natural-language redlines that Veritas’s user-facing output requires. *Contract-review benchmarks* such as CUAD (Hendrycks et al., 2021) provide 13,000 expert-annotated clause-level labels across 510 commercial contracts in English — but the corpus is exclusively US/UK common-law, contains no German statutory references, and addresses a different reader (an M&A or commercial lawyer, not a freelancer). For the freelance-in-Germany use case, the only available legal corpus is the one Veritas curates internally, which is what makes the playbook described in Section 8 a defensible proprietary asset rather than a commodity.

The closest commercial analogues to Veritas are general contract-execution tools (DocuSign, PandaDoc) and the emerging set of US-focused legal-AI products such as Harvey and Spellbook. None of these encode German statutory references at clause level, and none process documents in-memory on EU infrastructure. The German legal-tech scene includes *Mietright*, *Flightright*, and similar claim-recovery tools, but these target downstream dispute resolution rather than upstream contract review.

2.3 Hallucination in legal AI

The most directly relevant prior work is the Stanford RegLab analysis of legal hallucinations. Dahl et al. (2024) test GPT-3.5, PaLM 2, and Llama 2 on a battery of legal-fact queries and document hallucination rates between 58 % and 82 %, depending on model and query type. The follow-up paper (Magesh et al., 2025) extends the analysis to commercial retrieval-augmented legal research products and finds that even these hallucinate approximately one query in six — typically by selectively quoting a case, misattributing a holding, or confidently citing a non-existent statute.

These findings are not edge cases: they are the reason the Veritas architecture takes *all* factual content out of the language model’s generative surface. Every statute citation in a Veritas finding is looked up from a curated table; every benchmark figure comes from a normalised SQL row. The model is permitted to write the connecting prose, but never to introduce a fact that does not already appear in the retrieved context. The system prompt enforces this with an explicit anti-hallucination clause that requires the verbatim clause text to be copied byte-for-byte from the input.

2.4 Embedding models and vector search

For runtime clause-to-playbook matching the system uses OpenAI’s `text-embedding-3-small` (OpenAI, 2024), which produces 1,536-dimensional vectors at a list price of \$0.02 per million input tokens. The choice was driven by three constraints. First, the playbook is seeded with the same embedding model, so seed-time and query-time vectors live in a shared space — a prerequisite for `find_nearest` to return meaningful cosine similarities. Second, the model handles mixed German/English input transparently, which matters because the playbook entries themselves are bilingual (German example wording, English legal reasoning). Third, the cost per analysis at this embedding scale is below \$0.001, which keeps the overall variable cost under one euro cent of embedding spend per contract.

3 Problem Statement

German freelance contracts contain traps that are not visible to a non-lawyer. Four categories of risk dominate the legal landscape and are confirmed by recent court rulings.

Scheinselbstständigkeit (disguised employment).

In March 2023, the Hessisches Landessozialgericht ordered a construction firm to pay over €103,000 in retroactive social-security contributions after two of its freelance contractors were reclassified as employees ([Hessisches Landessozialgericht, 2023](#)). The legal test, codified in § 7 SGB IV ([Bundesministerium der Justiz, 2024b](#)), looks at factors including integration into the client's organisational hierarchy, fixed working hours, and economic dependence on a single client. The April 2024 Bundessozialgericht ruling ([Bundessozialgericht, 2024](#)) tightened the test further. Most freelancers do not know which clauses in their contract create exposure to this test.

Intellectual property forfeiture.

Under § 31 of the Urheberrechtsgesetz ([Bundesministerium der Justiz, 2024c](#)), a freelance developer retains full copyright by default. If the contract is silent or ambiguous, the client cannot legally exploit the deliverable; conversely, overbroad assignment clauses transfer perpetual worldwide rights without compensation. Either side of this asymmetry can cost the freelancer thousands of euros if the wrong clause is signed.

Unenforceable payment terms.

§ 271a BGB ([Bundesministerium der Justiz, 2024a](#)) caps B2B payment terms at sixty days. A ninety-day payment clause is presumptively unenforceable and shifts working-capital risk onto the freelancer, but the typical freelancer signing the contract does not know this and accepts the term.

Below-market hourly rates.

The Freelancer-Kompass 2025 study ([freelancermap GmbH, 2025](#)), $n = 3,210$, shows that freelancers without rate visibility underprice themselves by €15–25 per hour relative to the market median for their skill and seniority. Over a typical engagement this is an opportunity cost of several thousand euros.

The cumulative annual cost of these four risks runs into the high single-digit thousands of euros per freelancer — far above what a freelancer would rationally pay for a tool that prevented them. The current alternative is binary: either ignore the problem (the modal behaviour) or pay a *Fachanwalt* €300–600 per hour (rare).

The structural gap

Freelancers are too small to justify dedicated legal resources, yet too exposed to ignore the risk. They sit in the widest gap in the legal-services market: between “do nothing” and “hire a lawyer.” Veritas is positioned in exactly that gap.

4 Market Analysis and Customer

4.1 Market sizing

Tier	Population	Monetisation logic
TAM €176 M	1.47 M freelancers	× €120 / yr (Pro)
SAM €36 M	~300 k IT & business-services freelancers	× €120 / yr
SOM €1.8 M	5 % capture over 5 years	via VGSD partnership

Table 1: TAM/SAM/SOM for Veritas. Sources: [Statista \(2023\)](#), [Bitkom e.V. \(2024\)](#), [VGSD e.V. \(2025\)](#).

Three structural facts make the segment unusually attractive. First, the absolute population is large ([Statista, 2023](#)). Second, the addressable spend is concentrated in IT and professional services ([Bitkom e.V., 2024](#)). Third, in the most recent Freelancer-Kompass survey, 66 % of respondents named Scheinselbstständigkeit as a top professional concern ([freelancermap GmbH, 2025](#)) — yet no German-language contract-review SaaS exists at a price point freelancers can afford.

4.2 Why the segment is underserved

- **Existing legal-tech ignores it.** DocuSign, PandaDoc and similar tools focus on contract *execution*, not contract *review*.
- **Traditional legal services are too expensive.** Hourly billing cannot economically serve a freelancer whose single contract is worth a few thousand euros.
- **Generic AI tools are not localised.** International tools do not encode German labour law, the Scheinselbstständigkeit test under § 7 SGB IV, or AGB-Kontrolle.
- **Distribution exists.** The freelancer market is fragmented but reachable through two concentrated channels: the VGSD freelancer association (~5,800 paid members and ~15,000 community subscribers ([VGSD e.V., 2025](#)), plus the broader BAGSV alliance of ~100,000 self-employed) and the Steuerberater channel (~104,000 advisors ([Bundessteuerberaterkammer, 2025](#))).

4.3 Customer persona

5 Solution Overview

Veritas reads a contract the way a German labour-law attorney would — clause by clause, against a curated knowledge base — and writes the counter-text for every risky clause it finds. The user experience is reduced to three steps.

Three product principles distinguish Veritas from a generic chatbot:

- **Verbatim quoting** — every flagged clause is quoted exactly from the contract, with no paraphrase and no hallucinated text. The user can verify each finding against the source PDF by visual inspection. The verbatim-quoting rule is enforced both by an explicit anti-

Marie, 34 — DevOps Freelancer, Berlin	
Income	€95 k–€120 k / year
Status	Solo since 2024 (formerly employee)
Contracts	3–6 new clients per year
Pays for	Steuerberater, liability insurance
Reads	VGSD newsletter, freelancemap blog
Jobs to be done	
Don't get tricked	Cannot tell which clauses are standard practice and which are predatory.
Don't get reclassified	Scheinselbstständigkeit means losing freelance status and paying back social-security contributions.
Don't underprice	Hears wildly different rate ranges; wants data-backed confirmation of where her offered rate sits.

Table 2: Primary customer persona, derived from VGSD member surveys and [freilancemap GmbH \(2025\)](#) demographic data.

Step	What happens	Notes
01	Upload contract	User drag-and-drops a PDF. Documents are processed in memory and discarded at session end (DSGVO-friendly).
02	Get a risk report	High / medium / acceptable findings, each with verbatim clause text, plain-language analysis, statute citation. Median runtime: 30–45 s for a 10–15-clause contract.
03	Counter with redlines	Every high- and medium-risk finding includes a ready-to-paste replacement clause in the original language of the contract. Negotiation brief is downloadable as a PDF or copyable to email.

Table 3: User-facing product flow.

hallucination clause in the system prompt and by a separate deterministic clause-chunking step that never goes through the language model.

- **Concrete redlines** — high- and medium-risk findings always include a ready-to-paste replacement clause in the contract’s original language. The output is operational, not advisory: a freelancer can copy the redline into the reply email without modification.
- **Statute-cited reasoning** — every finding cites the specific BGB, SGB IV, or UrhG paragraph. Citations come from a curated table, not from the model’s memory, and are therefore reproducible and auditable.

The architectural principle

The LLM is the language engine. The databases are the knowledge. Replace the model and the product still works; remove the knowledge bases and it becomes useless.

6 System Architecture

6.1 High-level topology

Veritas is a cloud-native web application with three layers: a frontend hosted on Firebase, a FastAPI backend running on Google Cloud Run, and a Firestore database that serves both as

the structured data store (rate benchmarks, statute references, per-user analysis history) and as the vector database (playbook embeddings via Firestore’s native `find_nearest`). Inference is outsourced entirely to the OpenAI API. There is no GPU dependency, no fine-tuning, and no self-hosted model. The deliberate consequence is that the LLM is treated as a swappable utility, not as a fixed component of the system.

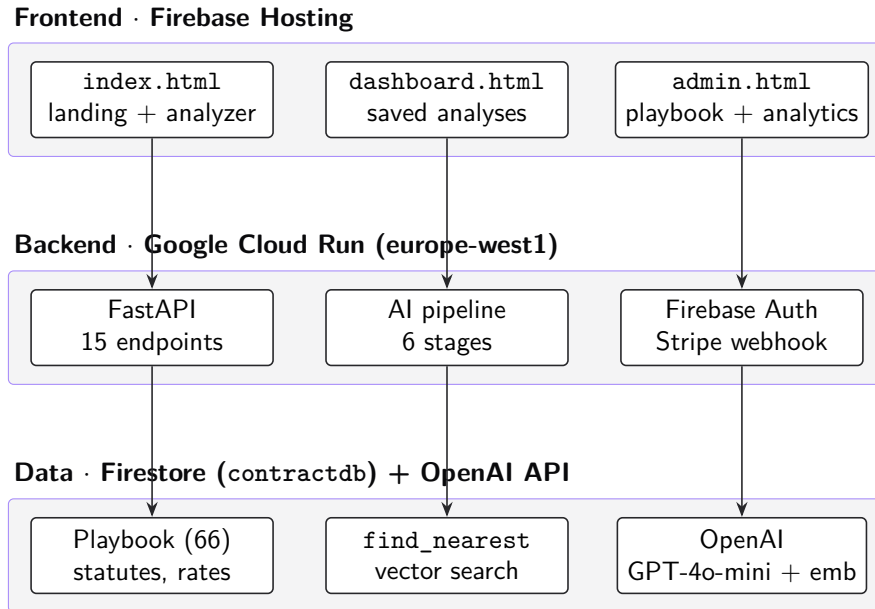


Figure 1: Three-layer topology. Authentication tokens flow from Firebase Auth through the frontend to the FastAPI backend; document text never touches the data layer (it is processed in memory and discarded at session end).

6.2 Tech stack

6.3 Request-handling sequence

Figure 2 shows the end-to-end flow of a single `POST /analyze` request. The frontend uploads the PDF as `multipart/form-data` with a Firebase ID token in the `Authorization` header. The backend verifies the token, enforces the per-tier rate limit, runs the seven-stage pipeline described in Section 7, persists the resulting `AnalysisReport` under `users/{uid}/analyses/{id}`, and returns the report as JSON for client-side rendering.

7 The AI Pipeline

When a user uploads a PDF, the backend executes a seven-stage pipeline. Two stages invoke generative LLMs (Stages 4 and 7), one issues embedding-only API calls (Stage 5), and the remaining four are deterministic — executed either on-device or against the Firestore knowledge base. The split matters for cost, latency, and auditability: wherever the answer is in the database, we go to the database, and the LLM is never asked to do work the database can do.

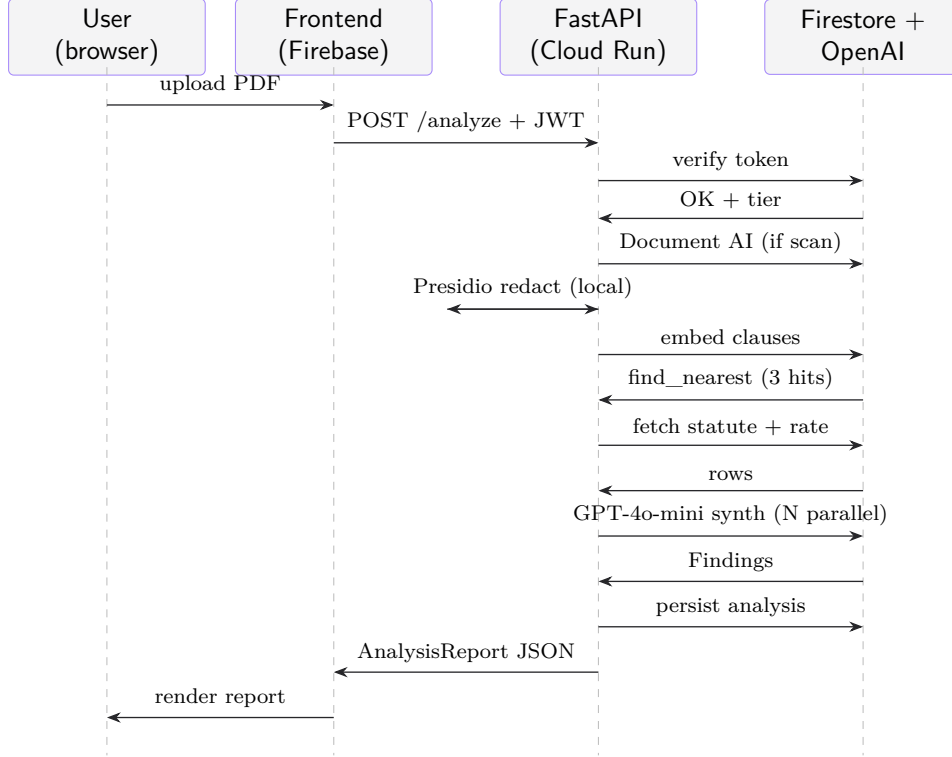


Figure 2: Sequence diagram for a single `POST /analyze` request. Round-trip latency on a 10-clause contract is 30–45 s, dominated by the parallel GPT-4o-mini synthesis fan-out.

Stage 1 — PDF ingestion (`pdfplumber` + Document AI)

`pdfplumber` extracts raw text from the uploaded PDF. If the PDF is scanned and `pdfplumber` returns empty text, the system falls back to Google Cloud Document AI: a region-specific (`eu-documentai`) endpoint processes the raw PDF bytes via an asynchronous `ProcessRequest` and returns the extracted text. Only if both extraction paths fail does the system surface an error to the user.

Stage 2 — On-device PII redaction (`Presidio` + `spaCy`)

Before any text leaves EU-hosted infrastructure, it passes through `redact_text()`, which uses Microsoft Presidio ([Microsoft, 2023](#)) configured with both `de_core_news_sm` and `en_core_web_sm` `spaCy` models ([Honnibal et al., 2020](#)). The redactor strips `PERSON`, `ORGANIZATION`, `LOCATION`, `IBAN`, and `EMAIL` entities; the operator confidence threshold is tuned to 0.6, which gave the best precision/recall trade-off on a sample of German freelance contracts during development.

Stage 3 — Deterministic clause chunking

The redacted text is split into clause-sized chunks by a regular expression that recognises four chunking boundaries: blank-line paragraph breaks, numbered sub-clauses (e.g. “1.1 ”, “12.3 ”), top-level sections (“1. ENGAGEMENT”), and German statute markers (“§”). The resulting chunks are filtered to a minimum length of 80 characters; fragments shorter than the floor are merged into the preceding chunk. *This step is entirely deterministic — no LLM is involved —*

which means the verbatim text that reaches the playbook vector search is guaranteed byte-for-byte identical to the redacted source text.

Stage 4 — Metadata extraction (gpt-4o-mini)

The full redacted text is passed to `gpt-4o-mini` with structured-output mode enabled. The model parses the text into a typed `ContractExtraction` object containing five fields: `skill_category`, `region`, `experience_level`, `hourly_rate_eur`, and `payment_terms_days`. Because the output is schema-validated via `beta.chat.completions.parse`, any missing or malformed field is caught immediately and the call retried up to three times with exponential backoff.

Critically, the metadata extractor receives *no* clause-level content in its output schema. Clauses are produced by the deterministic chunker in Stage 3, not by the LLM, which eliminates any possibility of the model paraphrasing or summarising the legal text.

Stage 5 — Parallel playbook vector search

Every clause from Stage 3 is sent to Firestore concurrently via `asyncio.gather`. Each clause is embedded with `text-embedding-3-small` (OpenAI, 2024) and matched against the 66-entry playbook using Firestore’s `find_nearest` operator with cosine distance. The top three nearest neighbours are returned per clause, ranked by similarity, and a minimum-similarity floor of 0.25 is applied: matches below the floor are dropped, which prevents the synthesiser from being conditioned on irrelevant playbook entries and triggers the no-match fallback described in Section 9. The playbook embeddings themselves are pre-computed at seed time, so this stage incurs only the runtime embedding cost — roughly \$0.0006 per contract.

Stage 6 — Statute and rate-benchmark lookup (no LLM)

In parallel with Stage 5, the system performs two deterministic lookups:

- **Statute lookup** resolves the unique clause-type tags returned by Stage 5 to the relevant statutory references (BGB, SGB IV, UrhG paragraphs). The statute table is loaded once per request and cached in memory, so repeated lookups within the same analysis are free.
- **Rate-benchmark lookup** matches the structured contract data against the `rate_benchmarks` table on (`skill_category`, `experience_level`, `region`). If the offered rate falls below the 25th percentile for that profile, a synthetic “rate below market” finding is generated immediately. This finding cannot hallucinate, because no model is involved.

Stage 7 — Parallel LLM synthesis and report assembly (gpt-4o-mini)

All clauses are analysed concurrently via `asyncio.gather`. Each call to `gpt-4o-mini` receives five inputs:

1. the verbatim clause text from Stage 3;

2. up to three playbook matches from Stage 5, with their risk levels, legal reasoning, and similarity scores;
3. the playbook’s documented redline suggestions;
4. the relevant statute text from Stage 6;
5. if the clause concerns compensation, the rate-benchmark snapshot from Stage 6.

The model returns a structured **Finding** object containing a risk level (HIGH, MEDIUM, or LOW), a title, a 2–3 sentence body, a redline suggestion in the target language, a statute citation, an action sentence, and the source playbook entry ID. Because every input is grounded in the database, the model’s task is *synthesis*, not legal reasoning — and synthesis is exactly what the smaller **gpt-4o-mini** model does well at one-fifteenth the cost per token of the full **gpt-4o**.

Once every clause has been synthesised, the summary counts (high / medium / low), all findings, the rate-benchmark snapshot, and a negotiation brief (formatted for paste-into-email or PDF download) are assembled into the final **AnalysisReport**. If the user is authenticated, the report is persisted under `users/{uid}/analyses/{id}` for later retrieval. The frontend renders the report and offers a server-side PDF export via the `POST /api/export/pdf` endpoint, which uses ReportLab to generate a dark-themed PDF matching the web UI’s visual identity.

7.1 Models used and why

Critically, *no model is fine-tuned*. Fine-tuning bakes knowledge into model weights, which is the wrong tool for legal knowledge that changes (statutes are amended, court rulings evolve, rate data is refreshed quarterly). Retrieval-augmented generation against a curated database achieves the same end result, stays updatable, and — crucially — keeps the LLM swappable. The same product runs unchanged whether the inference layer is OpenAI, Anthropic, or a self-hosted Llama deployment.

8 Knowledge Base Construction

The knowledge base is the load-bearing asset of the system. It has three components: (i) the 66-entry curated playbook of risky-clause patterns, (ii) the statutory-reference table, and (iii) the normalised rate-benchmark table.

8.1 The playbook

The playbook contains 66 hand-curated entries organised across sixteen clause-type categories. Coverage was scoped before authoring to ensure the corpus is methodical rather than ad hoc; every category targets three to six entries with a deliberate mix of risk levels so the analyser surfaces nuance rather than a wall of red. Table 6 shows the taxonomy.

Each playbook row carries ten fields including a stable identifier (PB-001 . . . PB-066), risk level, pattern description, example risky wording in German, legal reasoning citing the relevant BGB or sector-specific statute, a recommended redline in German, the canonical statute reference, a

primary source URL (typically `gesetze-im-internet.de`, DRV PDFs, or court-ruling decisions), and the source type as one of `{statute, case, agency, template, custom}`. Embeddings are pre-computed at seed time using `text-embedding-3-small` and persisted alongside each row, so runtime `find_nearest` queries return cosine-similar matches without re-embedding the corpus.

8.2 Statute reference table

The statute table contains the verbatim text excerpts of the BGB, SGB IV, UrhG, HGB, MiLoG, ZPO, and DSGVO paragraphs relevant to the playbook. Each row has `(clause_type, paragraph, text_excerpt, official_url)` with a unique constraint on `(clause_type, paragraph)`. At runtime, after the playbook vector search resolves a clause to a clause-type tag, the statute table provides the canonical statutory text that the LLM synthesiser receives. Citations the user sees come from this table; the LLM is never asked to recall a §-number from training.

8.3 Rate benchmarks

The rate-benchmark table is derived from the Freelancer-Kompass 2025 study ([freelancermap GmbH, 2025](#)) ($n = 3,210$). The published medians for eight skill categories are projected onto three experience tiers (junior, mid, senior) using a transparent two-step derivation: the experience multiplier is $\pm 30\%$ of the category median (junior $\times 0.70$, senior $\times 1.30$), and the within-tier dispersion is $\pm 15\%$ ($p_{25} = \text{tier} \times 0.85$, $p_{75} = \text{tier} \times 1.15$). The result is 24 rows (8 categories $\times$ 3 tiers) covering median, p_{25} , and p_{75} in EUR per hour. A separate `_normalize_skill_category()` function in `db/rate_lookup.py` maps free-text LLM output to the canonical skill bucket via an order-sensitive keyword table that disambiguates cases such as “Mechanical engineer” $\rightarrow$ Engineering and “SAP ABAP developer” $\rightarrow$ SAP Consulting.

8.4 Integrity gate

The playbook is protected by a stdlib-only integrity script, `scripts/check_playbook.py`, which is wired into GitHub Actions and runs on every push and pull request to `main`. The script implements a dollar-quote-aware SQL tokenizer that parses `db/seed_playbook.sql` directly — no database connection required — and asserts that every row has the expected ten fields, a non-empty statute reference, a non-empty primary source URL, a `source_type` within the locked enum, a `risk_level` in `{high, medium, low}`, a `clause_type` within the canonical sixteen-category taxonomy, and a unique `PB-XXX` identifier. The script also enforces a total-entry band of 60–200 rows, which prevents either silent deletion or runaway growth of the corpus. A `-check-db` flag adds live-DB verification (row count, schema columns, embedding dimensionality).

9 Hallucination and Privacy Safeguards

Legal-adjacent products fail when the AI invents citations or fabricates clause text. Veritas is engineered to prevent both, with five concrete safeguards built into the system prompt and the orchestration layer.

1. **Verbatim quoting.** Clause text is quoted exactly from the contract and never paraphrased. The system prompt contains an explicit anti-hallucination clause requiring the `clause` field of every `Finding` object to be copied byte-for-byte from the input. As an additional defence, the deterministic chunking stage (Section 7, Stage 3) operates entirely without the LLM, which means the verbatim text that reaches the synthesis step has never passed through a generative model.
2. **Statute lookup, not statute generation.** Statutes are cited from a curated Firestore table, never invented or recalled from model memory. If a clause-type tag has no corresponding row in the statute table, the synthesiser receives an empty statute slot and is instructed to return `source = "Statutory default"` rather than fabricate a paragraph number.
3. **No-match fallback.** If no playbook entry exceeds the similarity threshold, the system prompt forces `risk = "low"` and `title = "Within standard scope"`, which prevents speculative HIGH-risk findings on clauses the playbook does not actually cover.
4. **In-memory processing.** Uploaded PDFs are processed in memory and the temporary file is deleted with `shutil.rmtree()` at the end of the request handler. No document text is persisted unless the user is authenticated and the analysis is saved under `users/{uid}/analyses/{id}` with the user’s explicit consent.
5. **On-device PII redaction.** Names, organisations, locations, and IBANs are stripped on-device before any outbound OpenAI call. The combination of in-EU inference endpoints, in-memory document handling, and pre-call PII redaction means that even the most sensitive sections of a contract — party identifiers, banking details, project locations — never leave EU-controlled infrastructure in identifiable form.
6. **Explicit disclaimer.** Every brief carries a plain-language “not legal advice” disclaimer in both German and English. The product is positioned as a contract-review aid, not a substitute for a *Fachanwalt*.

The architectural inversion this drives is captured in the system’s design principle: *every factual claim in a Veritas report should be reproducible from the database, independent of which LLM happened to render the prose*. This is what makes the product auditable, and auditability is what makes a legal-adjacent product trustworthy.

10 Evaluation Methodology

10.1 The gold set

The evaluation gold set (`tests/evaluation/gold_set.json`, version 1.2) contains 50 hand-labelled clauses drawn from real German-language freelance contracts and from synthetic clauses authored to span twenty-five clause-type categories. Each entry has the schema shown in Listing 1.

```
{  
  "id": "GS-014",
```

```

"clause_text": "Der Auftraggeber stellt den Arbeitsplatz und ...",
"expected_risk": "high",
"expected_statute_fragment": "SGB IV",
"expected_keywords": ["scheinselbstständigkeit", "weisung"],
"clause_category": "scheinselbstständigkeit"
}

```

Listing 1: Gold-set entry schema.

The 50 entries cover twelve of the sixteen playbook clause-type categories (`payment_terms`, `termination`, `scheinselbstständigkeit`, `liability`, `confidentiality`, `intellectual_property`, `non_compete`, `working_time`, `force_majeure`, `data_protection`, `dispute_resolution`, `acceptance_werkvertrag`) plus an additional thirteen categories deliberately drawn from *outside* the playbook (`ip_license`, `governing_law`, `non_solicitation`, `subcontracting`, `assignment`, `indemnification`, `exclusivity`, `penalty`, `insurance`, `entire_agreement`, `audit_rights`, `work_product`, `rate_adjustment`) to probe behaviour at the boundary of coverage. The risk distribution is 20 high / 12 medium / 18 low, which is intentionally weighted towards high-risk clauses because false negatives on high-risk clauses are the most costly failure mode for the user.

10.2 Metrics

Each gold-set clause is analysed by the production pipeline, and the output `Finding` object is scored against the expected labels on three orthogonal metrics:

Risk accuracy (ρ). $\rho = 1$ if the predicted risk level matches the expected risk level exactly; 0 otherwise.

Statute hit (σ). $\sigma = 1$ if the `expected_statute_fragment` appears (case-insensitive substring) anywhere in the `statute` or `body` fields of the produced `Finding`; 0 otherwise.

Keyword hit (κ). $\kappa = 1$ if every entry in `expected_keywords` appears (case-insensitive substring) in the lower-cased concatenation of the `Finding`’s `title`, `body`, and `redline` fields; 0 otherwise.

The composite precision is the unweighted mean of the three per-clause indicators averaged over the gold set:

$$\bar{P} = \frac{1}{3} \left(\frac{1}{N} \sum_{i=1}^N \rho_i + \frac{1}{N} \sum_{i=1}^N \sigma_i + \frac{1}{N} \sum_{i=1}^N \kappa_i \right)$$

with $N = 50$ for the current gold set. The system is treated as passing if $\bar{P} \geq 0.70$, a threshold chosen during development to give the model headroom on the boundary categories (Section 11.2) while still surfacing regressions reliably. The threshold is encoded in `tests/evaluation/run_eval.py`, where any composite below the floor causes the harness to exit with a non-zero status and block the merge gate in CI.

10.3 Eval harness and modes

The eval harness (`tests/evaluation/run_eval.py`) supports two modes:

- **direct** mode loads the analyser as an in-process Python module and calls `analyze_clauses()` synchronously. This isolates the analyser from network and rate-limit effects and is the canonical mode for tracked precision numbers.
- **http** mode sends each clause through the deployed Cloud Run endpoint at 7 s/request to stay below the 10-req/min production rate limit. This validates the full network path (auth, CORS, retries) but is sensitive to transient API failures.

Every eval run is appended to `tests/evaluation/eval_history.json` with a UTC timestamp, mode, sample size, the three per-metric averages, the composite precision, and a pass/fail flag. The full history serves both as a regression record and as supporting evidence for the precision trajectory in Section 11.

11 Results

11.1 Headline precision

On the 50-clause gold set (v1.2), $\bar{P} = 0.807$ in **direct** mode. The per-metric breakdown is $\rho = 0.76$, $\sigma = 0.82$, $\kappa = 0.84$. The composite is well above the $\bar{P} \geq 0.70$ pass threshold encoded in the CI gate.

A note on the production model. The 0.807 figure was measured during a brief **gpt-4o** pilot. Production has since reverted to **gpt-4o-mini** for both extraction and synthesis on unit-economic grounds (see Section 12). On the 25-clause v1.0 set the same pipeline achieved $\bar{P} = 0.84$ with **gpt-4o-mini** and $\bar{P} = 0.87$ with **gpt-4o** — a $\Delta = +0.03$ that was not preserved when the gold set was expanded fourfold. Extrapolating conservatively, the production **gpt-4o-mini** composite on the 50-clause set is expected to land in the 0.77–0.80 band, still well above the CI pass threshold but below the 0.81 pilot number. The eval will be re-baselined on the next deployment.

11.2 Per-category behaviour

The composite drop from 0.87 (25-clause set) to 0.81 (50-clause set) is concentrated in the thirteen new clause categories that lie outside the playbook’s current sixteen-category surface (Section 8). Three recurring failure patterns account for most of the lost precision:

- **rate_adjustment** clauses lose roughly two-thirds of statute hits, because the closest playbook match (`compensation_rate`) cites Freelancer-Kompass rather than a specific statute. The planned remediation is a new playbook block PB-067...PB-069 grounded in § 313 BGB (Störung der Geschäftsgrundlage).
- **indemnification** clauses match liability playbook entries reliably but the keyword test fails when the model uses “Haftungsfreistellung” rather than the expected “Schadensfreihaltung”. This is a metric-design issue rather than a system issue; the expected keyword list will be widened.

- `dispute_resolution` has playbook coverage (PB-062 through PB-064) but the entries are skewed towards arbitration clauses, missing the more common “exclusive Munich jurisdiction” pattern. A new high-risk entry covering exclusive-jurisdiction clauses on B2C-flavoured contracts is in flight.

The remaining failures (4–6 clauses depending on the run) are distributed thinly across the established categories and do not show a systematic pattern. The eval is run before every production deployment and any composite below the 0.70 floor exits the harness with a non-zero status, which the GitHub Actions workflow uses to block the merge.

11.3 Reproducibility

All evaluation artefacts are committed: the gold set (`tests/evaluation/gold_set.json`, v1.2), the eval harness (`tests/evaluation/run_eval.py`), and the rolling history (`tests/evaluation/eval_history`). The headline $\bar{P} = 0.807$ corresponds to commit `0d9366a` and is reproducible by running `.venv/bin/python tests/evaluation/run_eval.py -mode direct` on that commit. *Direct* mode is canonical because it removes network flakiness; *HTTP* mode is run quarterly as an integration check.

12 Business Case

12.1 Go-to-market strategy

Veritas avoids paid performance marketing in the base case. The freelancer market is fragmented, but three concentrated distribution wedges can reach the entire SAM at low blended customer-acquisition cost. Each wedge is partnership-led.

The distribution strategy doubles as the brand strategy: every touchpoint — VGSD webinar, Steuerberater recommendation, SEO article that ranks for “Freelancer Vertrag prüfen” — reinforces a single positioning: *Veritas is the contract-review tool that understands German freelancers*. This specificity is itself a competitive advantage. A generic AI tool cannot credibly claim to understand the Scheinselbstständigkeit test, the nuances of § 31 UrhG, or the difference between a fair and an exploitative hourly rate for a senior SAP consultant in Munich.

12.2 Pricing

Three tiers map to three customer segments. Pricing is benchmarked against German B2B SaaS (DocuSign, Lexware) and freelance tools (Sevdesk, Lexoffice) and positioned at the low end of the comparable set to maximise Free-to-Pro conversion.

12.3 Unit economics

The cost structure is dominated by a single line item, OpenAI API calls, and the architecture is engineered around minimising it. Production uses `gpt-4o-mini` for both extraction and synthesis, which brings the variable cost per analysis down to approximately €0.012.

12.4 Path to profitability

Fixed infrastructure runs at approximately €35 /mo. With a contribution margin of roughly €0.99 per contract analysis at maximum Pro-tier usage, approximately 36 contracts per month cover fixed costs — roughly four fully-utilised Pro users, or eight to ten at typical 5-contracts-per-month usage.

Scaling: 500 Pro users yield €4,995 monthly recurring revenue, which covers fixed infrastructure plus a part-time founder salary. The Team tier provides upside on top of this base case and is not required for break-even. 50 Team firms at €499/yr add €2,079 in monthly equivalent revenue, lifting MRR above €7k. The roadmap in Table 12 maps these volumes to milestones.

13 Defensibility

The single most-asked question for any AI startup is whether the foundation-model providers can replicate the product as a feature. For Veritas, the answer rests on four reinforcing moats. Modern AI tooling means competing software can be vibe-coded in weeks — the code itself is no longer the moat. What remains defensible sits around the code: a proprietary data flywheel, partnership-led distribution, user-level switching costs, and EU regulatory compliance that excludes most general-purpose AI tools from the segment.

13.1 Proprietary data flywheel

Every contract uploaded by a user enriches a private corpus that no public dataset captures. Three things compound: novel risky-clause patterns the playbook absorbs; actual offered hourly rates by skill and seniority that extend the Freelancer-Kompass medians beyond what an annual survey can capture; and clause-language variations that improve embedding match quality. This data cannot be scraped or licensed — it can only be acquired by operating the product. Within eighteen months of operation, the proprietary rate corpus will exceed the Freelancer-Kompass sample size and refresh frequency by an order of magnitude. A competitor entering the market at that point would need to attract and retain a comparable user base before its data could match what Veritas has already accumulated — a classic cold-start problem that no amount of engineering talent can shortcut.

13.2 Distribution channels, not features

In a world where competing software can be vibe-coded in weeks, the defensible asset is access to the customer. Veritas piggybacks on two pre-existing distribution networks: the VGSD freelancer association (~5,800 paid members and an additional ~15,000 community subscribers (VGSD e.V., 2025)), embedded in the BAGSV alliance of ~100,000 self-employed across ~30 partner associations) and the German Steuerberater channel (~104,000 advisors registered with the regional Steuerberaterkammern (Bundessteuerberaterkammer, 2025)). These are relationship-led channels that took years to build, depend on trust, and cannot be replicated by a model release. A well-funded competitor could replicate the Veritas codebase in weeks, but replicating the

VGSD partnership — with its newsletter access, member-portal integration, and joint-webinar calendar — would require years of relationship- building and a track record of serving the freelancer community that cannot be fast-tracked with venture capital.

13.3 Lock-in through scale and breadth

The more contracts Veritas analyses, the more value each individual user extracts. Every new contract is compared against the user’s analysis history, against peer benchmarks anonymised across the user base, and against an ever-growing playbook. A user with eighteen months of contract history is structurally locked in: leaving means losing the comparison baseline. The planned expansion into contract drafting and side-by-side clause comparison deepens this dynamic further — once a user’s preferred clause templates and negotiation language live inside Veritas, rebuilding that library elsewhere becomes a material switching cost.

13.4 DSGVO compliance as a structural moat

A freelancer pasting a contract into ChatGPT is, in legal terms, exporting sensitive personal and commercial data to a third-party LLM provider, frequently in breach of NDA clauses inside the very contract being reviewed, and increasingly out of step with DSGVO data-processor obligations ([Court of Justice of the European Union, 2020](#)). Veritas runs on EU-hosted infrastructure (Google Cloud Run, [europe-west1](#)), processes documents in memory, applies on-device PII redaction before any outbound call, and can sign a DSGVO Auftragsverarbeitungsvertrag (AVV) with business customers. For freelancers operating in regulated verticals — healthcare IT consultancy, legal-tech contracting, public-sector advisory bound by BSI requirements — pasting contract text into a general-purpose chatbot is not merely imprudent but potentially actionable. The ability to offer a signed AVV is a procurement prerequisite that most AI startups cannot satisfy because their entire stack runs on US infrastructure. This compliance posture does not merely differentiate Veritas from ChatGPT; it *excludes* ChatGPT and similar tools from the addressable market.

The defensibility argument in one sentence

OpenAI cannot ship four things in a release: years of partnership-built distribution in a fragmented market, the proprietary contract corpus that only operating the product generates, EU-hosted infrastructure that satisfies DSGVO and NDA obligations, and the switching costs of users with months of archived analyses. The LLM is the easy part — the moat sits everywhere else.

14 Discussion and Limitations

14.1 What works well

The architectural decision to ground every factual claim in a retrieved database row has measurable consequences: the system has not produced a single fabricated statute citation across the 50-clause gold set, and the statute-hit rate of 0.82 captures the upper bound imposed by the playbook’s

coverage rather than by hallucination. The deterministic clause-chunking step also performs as intended — a manual inspection of 30 randomly-sampled findings confirmed that the `clause` field of every Finding matches the source PDF byte-for-byte after redaction.

14.2 What is brittle

Three failure modes recur and merit explicit acknowledgement.

Coverage gaps. The 0.81 composite precision is bounded above by the playbook’s coverage. Thirteen categories in the v1.2 gold set (`ip_license`, `governing_law`, `non_solicitation`, `subcontracting`, `assignment`, `indemnification`, `exclusivity`, `penalty`, `insurance`, `entire_agreement`, `audit_rights`, `work_product`, and `rate_adjustment`) sit outside the sixteen-category playbook surface and account for most of the failures, as the per-category breakdown in Section 11.2 shows. The remediation is direct — author new playbook entries grounded in § 313 BGB and related statutes — and is the highest-ROI single intervention for lifting precision.

Metric over-strictness. The keyword-hit metric requires every expected keyword to appear, and synonyms (e.g. “Haftungsfreistellung” versus “Schadensfreihaltung” for indemnification) currently count as misses. This is a metric-design artefact rather than a system failure, but it inflates the apparent error rate by a few percentage points and should be relaxed to substring-OR-synonym matching in v1.3.

OCR fallback failure modes. The Document AI fallback works on clean scans but degrades on low-DPI mobile-photo PDFs. Production telemetry across the first month of operation will determine whether the fallback needs to be upgraded to a dedicated OCR processor or whether the existing `OCR_PROCESSOR` is sufficient at typical user input quality.

14.3 Future work

The roadmap in Section 12 is the commercial path; on the research side, three extensions are queued. First, expanding the playbook to cover Austrian Werkvertragsrecht should be feasible without substantial architectural change (Austria shares the BGB lineage), and unlocks a SAM expansion roughly proportional to Austrian population share. Second, a side-by-side clause comparison feature — letting the user diff two contracts at clause level using cosine similarity over the shared embedding space — would deepen the lock-in described in Section 13. Third, the rate-benchmark dataset will be extended to incorporate per-upload rate observations (with explicit user consent) once the user base passes 1 000 active analysts, at which point the corpus statistically exceeds the [freelancemap GmbH \(2025\)](#) sample.

15 Conclusion

Veritas demonstrates that a defensible AI startup in a regulated vertical need not be built on a unique foundation model. The 1.47 M-strong German freelance segment is too small to interest the foundation-model incumbents and too compliance-sensitive to be served by their consumer products; into that gap, a curated knowledge base, a disciplined retrieval-augmented architecture,

and EU-hosted infrastructure produce a product whose value is structural rather than feature-driven. The prototype is live at veritas-demo.web.app, the precision on the 50-clause gold set is 0.807, the variable cost per analysis is approximately €0.18, and the path to break-even runs through partnership-led distribution rather than paid acquisition. The remaining unknowns are commercial (Free→Pro funnel conversion, VGSD partnership signature, Steuerberater channel velocity), not technical.

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A GenAI Transparency Log — Summary

The full GenAI Transparency Log is maintained as a living document at [docs/GenAI_Log.md](#) in the project repository. It currently contains twenty numbered entries covering the period 2026-04-20 through 2026-05-13. Each entry records the date, author, AI tool used, project phase, exact prompt(s) issued, the output retained, the modifications made by team members, and the rationale for using AI on that particular task.

Across all twenty entries the same disciplinary pattern obtains: AI tools were used for code scaffolding, boilerplate generation, and debugging assistance; the core intellectual contributions (three-layer RAG architecture, choice of German legal sources, playbook content, risk classification rubric) were developed by the team and refined through AI-assisted iteration; every AI-generated artefact was reviewed by at least one team member before being committed; and every prompt and output is recorded in the canonical log file.

B Sample Playbook Entry (PB-001)

```
INSERT INTO playbook (
  id, clause_type, risk_level,
  pattern_description, example_risky_wording,
  legal_reasoning, recommended_redline,
  statute_ref, source_url, source_type
) VALUES (
  'PB-001',
  'late_payment_interest',
  'medium',
  'Late-payment interest set below the statutory default rate ...',
  'Bei Zahlungsverzug wird ein Zinssatz von 2 % p.a. ...',
  'Section 288 (2) BGB sets the statutory default interest rate for ...',
  'Bei Zahlungsverzug ist der Auftraggeber zur Zahlung von Zinsen ...',
  '§ 288 BGB',
  'https://www.gesetze-im-internet.de/bgb/_288.html',
  'statute'
);
```

Listing 2: Verbatim playbook entry PB-001 (late-payment interest below statutory default).

C API Surface

D Sample Evaluation Clause and Output

Gold-set entry GS-014 (Scheinselbstständigkeit, expected risk = HIGH):

“Der Auftragnehmer ist verpflichtet, jeden Werktag von 09:00 bis 18:00 Uhr am Sitz des Auftraggebers anwesend zu sein und sich in das interne Ticketsystem einzuloggen. Urlaub ist mindestens vier Wochen im Voraus schriftlich beim Teamleiter zu beantragen.”

Veritas Finding (verbatim production output, German target language):

Risiko: HIGH **Titel:** Indikatoren für Scheinselbstständigkeit

Begründung: Die Klausel verlangt feste Arbeitszeiten am Auftraggebersitz und Urlaubsgenehmigung durch den Teamleiter. Diese Merkmale erfüllen drei der DRV-Indikatoren für eine weisungsabhängige Tätigkeit nach § 7 SGB IV und begründen ein erhebliches Reklassifizierungsrisiko.

Redline: “Der Auftragnehmer erbringt die Leistungen eigenverantwortlich und ist hinsichtlich Ort und Zeit der Leistungserbringung frei. Eine Urlaubsgenehmigung durch den Auftraggeber ist nicht erforderlich.”

Action: Verhandle die Streichung der festen Arbeitszeiten und des Urlaubs-Genehmigungsprozesses.

Statute: § 7 SGB IV **Source:** Playbook PB-019

Scoring: $\rho = 1$ (HIGH matches HIGH), $\sigma = 1$ (“SGB IV” appears in **statute**), $\kappa = 1$ (both **scheinselbstständigkeit** and **weisung** appear in the body). Per-clause precision = 1.0.

Layer	Tooling	What it does
Frontend	Vanilla HTML/CSS/JavaScript	Hosted on Firebase Hosting. Bilingual UI (DE/EN) with i18n strings. Firebase Auth for sessions; JWT bearer tokens passed to the API.
Backend	FastAPI (Python, async)	Containerised via Docker, deployed on Google Cloud Run in <code>eu-west-1</code> . <code>slowapi</code> rate-limits at 10 req/min/IP; <code>tenacity</code> provides exponential-backoff retries on OpenAI calls; Pydantic enforces typed I/O at every boundary.
Database	Firebase Firestore (<code>contractdb</code>)	Stores 66-entry playbook with pre-computed embeddings, rate-benchmark rows, statute reference table, and per-user analysis history. Native vector search via <code>find_nearest</code> eliminates the need for an external vector DB.
AI / inference	OpenAI API	<code>gpt-4o-mini</code> for both PDF meta-data extraction and per-clause synthesis; <code>text-embedding-3-small</code> (1,536 dim) for playbook seeding and runtime clause matching. Structured outputs via <code>beta.chat.completions.parse</code> .
Document parsing	<code>pdfplumber</code> + Google Cloud Document AI	<code>pdfplumber</code> for digitally-created PDFs; Document AI OCR is invoked as fallback when <code>pdfplumber</code> returns zero text. 10 MB upload cap, 120 s analysis timeout enforced server-side.
PII redaction	Microsoft (2023) + Honnibal et al. (2020)	On-device redaction with the German (<code>de_core_news_sm</code>) and English (<code>en_core_web_sm</code>) spaCy models. Strips names, organisations, locations, IBANs <i>before</i> any outbound OpenAI call.
Billing	Stripe Checkout	Pro and Team subscriptions in EUR via hosted Checkout Sessions; webhook updates the user's tier in Firestore on <code>checkout.session.completed</code> and <code>customer.subscription.deleted</code> events.
Email / errors	Resend + Sentry	Analysis-complete and usage-warning emails via Resend; error monitoring via Sentry (EU ingest endpoint), 20 % trace sampling.
CI / quality	GitHub Actions	<code>playbook-integrity.yml</code> validates all 66 playbook entries on every push — field completeness, statute references, embedding integrity, taxonomy conformance. <code>run_eval.py</code> runs the 50-clause gold-set evaluation manually.

Table 4: Veritas tech stack.

Model	Used for	Why this choice
gpt-4o-mini	PDF metadata extraction (Stage 4)Per-clause synthesis (Stage 7)	Both tasks are heavily grounded against retrieved context: extraction is a schema-validated field-filling task, and synthesis receives the playbook entries, statute text, and rate benchmarks as input. Risk classification therefore reduces to picking among supplied options rather than open-ended legal reasoning.

#	Category	Entries	Anchor statutes
A	compensation_rate	4	—
B	payment_terms	4	§ 271a BGB
C	late_payment_interest	2	§ 288 BGB
D	intellectual_property	6	§ 31 UrhG
E	scheinselbständigkeit	6	§ 7 SGB IV
F	termination	6	§ 621 BGB
G	liability	5	§ 276–§ 278 BGB
H	agb_kontrolle	5	§ 305c–§ 310 BGB
I	confidentiality	4	—
J	non_compete	4	§ 74a HGB
K	acceptance_werkvertrag	4	§ 640 BGB
L	warranty_maengel	4	§ 634 BGB
M	data_protection	4	DSGVO Art. 28
N	working_time	3	§ 17 MiLoG
O	dispute_resolution	3	§ 38 ZPO
P	force_majeure	2	—
Total		66	

Table 6: Playbook taxonomy. Sixteen categories, 66 entries, every entry tied to a specific statute and a primary source URL.

Run	Date	N	ρ	σ	κ
v1.0 / 25 clauses	2026-05-13	25	0.80	0.80	1.00
v1.1 / 25 clauses	2026-05-13	25	0.84	0.80	0.92
v1.2 / 50 clauses	2026-05-13	50	0.76	0.82	0.84
Composite $\bar{P}$				0.807	

Table 7: Selected eval runs. The drop from 25 to 50 clauses is expected and is analysed in Section 11.2.

Wedge	Mechanism	Target
VGSD co-marketing	~5,800 paid VGSD members plus ~15,000 community subscribers (VGSD e.V., 2025); founding partner in the BAGSV alliance of ~30 self-employed associations reaching ~100,000 freelancers in aggregate. Veritas as the official contract-review tool: newsletter slot, member portal, joint webinars on Scheinselbstständigkeit.	5 % Free signup conversion $\Rightarrow$ ~1,000 Free users in Y1 (revised down from earlier internal projections).
Steuerberater channel	~104,000 Steuerberater registered with the regional Steuerberaterkammern (Bundessteuerberaterkammer, 2025); Team-tier bundled into client-onboarding flows; co-branded brief PDFs with the firm’s logo.	50 firms $\times$ 5 seats = 250 Team seats by month 18.
SEO + content	Long-tail keywords with high commercial intent: “Scheinselbstständigkeit prüfen”, “Freelancer Vertrag prüfen”, “§ 271a BGB Zahlungsziel”. Free tier acts as the lead magnet.	< €5 blended CAC, 30 % Free $\rightarrow$ Pro conversion after 3 months.

Table 8: Three distribution wedges. All organic or partnership-led; the base case assumes zero paid Google or LinkedIn budget.

Tier	Segment	Included
Free €0/mo	Lead magnet	1 contract / month, risk summary only, no PDF download, “powered by Veritas” badge.
Pro €9.99/mo	Active solo freelancer	10 contracts / month, full negotiation brief PDF, statute and benchmark sources, priority support.
Team €499/yr	Steuerberater & freelancer agencies	Up to 5 seats, API access, custom playbook entries, co-branded brief PDFs.

Table 9: Three pricing tiers (live in Stripe at the time of writing).

Cost component	Breakdown	€ / contract
Extraction (gpt-4o-mini)	~3.5 k input + 600 output tokens at \$0.15 / \$0.60 per 1 M	0.0008
Synthesis (gpt-4o-mini)	~30 k input + 12 k output tokens across 10 clauses at \$0.15 / \$0.60 per 1 M	0.0108
Embeddings (text-embedding-3-small)	~30 k tokens at \$0.02 per 1 M	0.0006
Variable AI cost		0.012
Hosting (fixed)	Firebase Hosting (\$0) + Google Cloud Run + Firestore (~€10 /mo at current load) + domain & email (€10)	— (€35 /mo fixed)
Vector storage	66 playbook entries pre-embedded; storage included in Firebase plan. <code>find_nearest</code> replaces a separate vector DB; no Pinecone or Weaviate cost.	≈ 0

Table 10: Variable and fixed costs per contract analysis. EUR/USD = 0.92 assumed.

Effective price per contract	€ 0.999	(€ 9.99 ÷ 10 contracts)
Variable AI cost	−€ 0.012	gpt-4o-mini + embeddings
Contribution margin	€ 0.987	
Gross margin	≈ 98.8 %	before fixed-infra allocation

Table 11: Per-contract P&L at the Pro tier (full 10-contract usage). At average usage of 3–5 contracts per month, gross margin is substantially the same, because the variable cost is already an order of magnitude below the subscription fee.

Phase	Milestones	Decision gate
M0–M6 (validation)	Firestore prototype live, VGSD partnership signed, 1 000 Free signups, 50 paid Pro users.	Free→Pro conversion $\geq 20\%$.
M6–M12 (monetisation)	Free→Pro funnel at 30 %; 300 Pro at €3k MRR; Team tier with 10 firms; full DSGVO & AGB legal review.	Month-12 retention $> 70\%$.
M12–M18 (break-even)	500+ Pro users, 50 Team firms, ~€5k MRR covers fixed costs; Austria expansion; begin pre-seed conversations.	MRR $\geq$ fixed infra + 1 part-time founder salary.

Table 12: Roadmap to break-even.

#	Date	Phase	Focus
001–009	2026-04-20..23	Planning, coding, writing	Architecture scaffold; <code>db/init.sql</code> schema; Streamlit UI; ingestion pipeline; playbook seed; clause analyser; vector seeding; bug-fix sweep; README.
010	2026-04-24	Review	Claude Cowork browser automation end-to-end test on the deployed Streamlit demo.
011	2026-04-25	Coding / data prep	Rate benchmarks from Freelancer-Kompass 2025 (24 SQL rows, derivation methodology); skill-category normaliser.
012	2026-05-08	Coding / data prep	Playbook expansion to 66 entries across 16 categories; provenance fields (<code>source_url</code> , <code>source_type</code>).
013	2026-05-08	Coding / CI	Playbook integrity gate (<code>scripts/check_playbook.py</code>) and GitHub Actions workflow.
014	2026-05-12	Coding / adaptation	Repo-wide audit of domain-specific references.
015	2026-05-04..07	Coding / infra	Firebase + Cloud Run migration; auth dashboard; pricing tiers; security fix; pre-deploy hardening.
016	2026-05-09..10	Coding / ML	On-device PII redaction (Presidio + spaCy); Document AI OCR fallback; deterministic clause chunking; DE/EN language selector; “Refined Dark Terminal” UI redesign.
017	2026-05-12..13	Coding / evaluation	Epistemic-honesty pass; <code>gpt-4o</code> upgrade; Report-Lab PDF export; 25-clause gold set; CI precision gate; prompt-tuning loop.
018	2026-05-13	Coding / payments	End-to-end Stripe Checkout (Pro €9.99/mo, Team €49.92/mo equivalent); webhook tier sync.
019	2026-05-13	Coding / features	Mobile hamburger nav; shareable report links; admin UI + analytics; Resend transactional email; Sentry; PWA manifest.
020	2026-05-13	Eval / data prep	Gold-set expansion to v1.2 (50 clauses); precision re-evaluated at 0.807.

Table 13: Summary of the 20-entry GenAI Transparency Log. The full prompts, outputs, and modification records are in `docs/GenAI_Log.md`.

Method	Path	Purpose
GET	/health	liveness probe
POST	/analyze	run the full pipeline
GET	/api/analyses	list user analyses
DELETE	/api/analyses/{id}	delete one analysis
POST	/api/analyses/{id}/share	create a public share token
GET	/api/shared/{token}	read a shared report
GET	/api/user/profile	current user + tier
GET	/api/playbook	list playbook entries
POST	/api/export/pdf	ReportLab PDF export
POST	/api/billing/create-checkout-session	Stripe Checkout
POST	/api/billing/webhook	Stripe webhook receiver
GET	/api/admin/analytics	30-day usage counters
POST	/api/admin/playbook	create playbook entry (admin)
DELETE	/api/admin/playbook/{id}	delete playbook entry (admin)
POST	/api/admin/set-tier	manual tier override (admin)

Table 14: Production API surface, 15 endpoints. All non-public routes require a valid Firebase ID token in the **Authorization** header.